

**MINISTRY OF EDUCATION AND TRAINING  
FPT UNIVERSITY**

**A TWO-TIER COGNITIVE ARCHITECTURE FOR  
AUTOMATED THREAT DETECTION AND RESPONSE  
USING AGENTIC AI**

by

Nguyen Duc Binh

A thesis submitted in conformity with the requirements  
for the degree of Master of Software Engineering

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Supervisor:  
Dr. Bui Van Hieu  
Dr. Dang Van Hieu

## Abstract

In the modern cybersecurity landscape, Security Operations Centers (SOCs) face acute alert fatigue as continuous telemetry volume far exceeds human analyst processing capacity. Traditional automation tools relying on static playbooks lack contextual reasoning, whereas feeding raw logs directly to Large Language Models (LLMs) suffers severe latency bottlenecks, prompt injection risks, and data sovereignty violations. This thesis proposes SENTINEL, a cognitive two-tier security architecture operating fully air-gapped. Tier 1 functions as a wire-speed filter to automatically discard baseline stream noise, while Tier 2 employs a stateful AI reasoning agent backed by MITRE ATT&CK and NIST guidance to handle complex threats behind cryptographic guardrails.

Empirical evaluation on benchmark intrusion datasets demonstrates SENTINEL's high efficiency. Tier 1 automatically offloads 92.2% of stream noise at sub-millisecond latency with throughput exceeding 2,500 flows/s. Tier 2 achieves a 67.33% technique mapping accuracy (via RRF) and outperforms the Single-Tier baseline in generating accurate security actions. Delimited Nonce encapsulation neutralizes 100% of prompt injection attacks, while HMAC-SHA256 hash-chaining guarantees audit trail non-repudiation, operating on a single 16 GB GPU under optimal resource consumption. SENTINEL proves that a two-tier hybrid architecture provides a trustworthy and effective paradigm for modern SOC defense.

## Acknowledgments

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I am deeply grateful to the Board of Management, the leadership of the FSB School of Business and Technology, and the faculty members of FPT University. Their dedicated teaching equipped me with strong theoretical and practical knowledge, while providing a professional, modern, and supportive research environment throughout my Master's studies.

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Finally, and most cherished of all, I dedicate my deepest emotional gratitude to my beloved family, especially my dear wife and our young child. Their silent sacrifices, infinite patience, unwavering understanding, and constant encouragement during the countless late nights spent conducting intensive system evaluations served as my solid pillar of strength, enabling me to overcome all obstacles and successfully bring this work to fruition.

While every effort has been made to conduct this research with utmost rigor and scientific dedication, limitations inevitably remain. I humbly welcome all constructive feedback and guidance from the thesis examination committee and fellow researchers to further refine and expand this work in the future.

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## List of Abbreviations

| Abbreviation | Full Description                                            |
|--------------|-------------------------------------------------------------|
| AI           | Artificial Intelligence                                     |
| APT          | Advanced Persistent Threat                                  |
| CTI          | Cyber Threat Intelligence                                   |
| DAPT         | Dynamic Advanced Persistent Threat                          |
| FAISS        | Facebook AI Similarity Search                               |
| FPR          | False Positive Rate                                         |
| GGUF         | GPT-Generated Unified Format                                |
| HITL         | Human-in-the-Loop                                           |
| HMAC         | Hash-based Message Authentication Code                      |
| LLM          | Large Language Model                                        |
| MITRE ATT&CK | MITRE Adversarial Tactics, Techniques, and Common Knowledge |
| NIST         | National Institute of Standards and Technology              |
| RAG          | Retrieval-Augmented Generation                              |
| RRF          | Reciprocal Rank Fusion                                      |
| SIEM         | Security Information and Event Management                   |
| SOAR         | Security Orchestration, Automation, and Response            |
| SOC          | Security Operations Center                                  |

# Chapter 1

## Introduction

This chapter presents the research context, central problem statement, and architectural foundation of the SENTINEL system—a Cognitive Two-Tier Architecture designed to automate threat detection and incident response within Security Operations Centers (SOCs). By critically analyzing the structural limitations of conventional deterministic defenses, the alert fatigue crisis, and the latency bottlenecks and adversarial security vulnerabilities inherent in Large Language Models (LLMs), this chapter establishes the research objectives, strategic research questions, scope, Design Science Research methodology, and core academic contributions of this thesis.

### 1.1 Background and Motivation

The expansion of enterprise IT infrastructure alongside the growing sophistication of multi-vector cyberattack campaigns poses severe operational challenges for security monitoring. Within modern Security Operations Centers (SOCs), Security Information and Event Management (SIEM) platforms are tasked with ingesting and aggregating millions of telemetry log events daily [2]. Conventional defense mechanisms rely heavily on perimeter filtration devices operating on static rulesets or fixed signatures. This approach exhibits fundamental limitations against novel evasion techniques and obfuscated attack variants. To minimize the risk of missing critical threats, security configurations frequently broaden sensitivity thresholds, resulting in hundreds of thousands of daily alerts with false positive rates exceeding 80% [4]. Consequently, security analysts suffer from cognitive exhaustion (alert fatigue), severely compromising their ability to identify genuine intrusion events. Concurrently, legacy Security Orchestration, Automation, and Response (SOAR) platforms operate on pre-scripted playbooks, completely lacking adaptive contextual reasoning capabilities [5].

Integrating Large Language Models (LLMs) offers a promising avenue toward automating security context synthesis and reasoning. Nevertheless, deploying LLMs directly into live

network telemetry streams presents three core technical barriers: First, the computational complexity of the self-attention mechanism causes inference latencies ranging from 4 to 26 seconds per event, incurring severe data pipeline congestion at wire-speed [18]. Second, the adversarial security risk stemming from log-substrate prompt injection attacks, wherein malicious instructions embedded in raw log fields manipulate the model’s internal reasoning flow [9]. Third, stringent data sovereignty regulations and Zero-Trust architecture mandates strictly prohibit exfiltrating internal security telemetry to public cloud AI services, requiring reasoning engines to operate entirely on-premises on bounded GPU hardware [31]. To resolve these challenges, this research proposes the SENTINEL architecture, synergizing an  $O(1)$  statistical filter and a LightGBM Machine Learning Gate at Tier 1 with an Agentic AI reasoning system at Tier 2, safeguarded by cryptographic nonces and an HMAC audit trail.

## 1.2 Research Objectives

The overarching objective of this thesis is to design, prototypically build, and empirically evaluate the SENTINEL architecture to automate network security threat detection, contextual reasoning, and incident response on local infrastructure, achieving optimal processing latency and cryptographic security.

The general objective is operationalized through three central research tasks: First, constructing a dual-stage stateless filtration pipeline at Tier 1 operating at wire-speed, combining Welford’s  $O(1)$  online statistical algorithm, a LightGBM Machine Learning Gate, and a Tier 1.75 Semantic Cache to classify and shed at least 80% of confident benign telemetry without consuming GPU compute resources. Second, designing a structural protection mechanism for the Tier 2 language model via Delimited Data Encapsulation using random cryptographic nonces, an output action guardrail, and an HMAC-SHA256 log audit chain to neutralize prompt injection attacks and ensure forensic non-repudiation. Third, developing a stateful reasoning engine based on a LangGraph agent state-graph coupled with persistent Threat Memory and a Dual-RAG retrieval system (combining FAISS vector search and BM25 sparse keyword matching) integrated with a STIX-aligned MITRE ATT&CK knowledge base, enabling transparent forensic justification generation, detecting and neutralizing hallucination via deterministic RAG guardrails, and facilitating multi-stage intrusion correlation.

### 1.3 Research Questions

The design and empirical evaluation process of this thesis are guided by three strategic research questions:

The first research question addresses pipeline efficiency and filtration economics: To what extent can the hybrid Tier 1 filtration pipeline (combining Welford  $O(1)$  statistics, LightGBM Machine Learning Gate, and Tier 1.75 Semantic Cache) achieve autonomous telemetry offload at wire-speed, and what overall response latency reduction is attained compared to a pure language model processing paradigm?

The second research question addresses AI security boundaries and forensic integrity: To what degree does Delimited Data Encapsulation using random cryptographic nonces coupled with deterministic action guardrails neutralize log-substrate prompt injection attacks, and how does an HMAC log audit chain protect forensic non-repudiation against log manipulation attempts?

The third research question addresses stateful agent reasoning and technical attribution: How effectively does a stateful Tier 2 AI agent equipped with persistent Threat Memory and Dual-RAG correlate complex intrusion events, accurately attribute MITRE ATT&CK technique codes, and produce transparent forensic justifications?

### 1.4 Scope and Object of Study

The primary subject of study is the SENTINEL Two-Tier Cognitive Architecture, comprising the Tier 1 stateless filter (online Welford statistics and LightGBM Machine Learning Gate), Tier 1.75 Semantic Cache, Tier 2 stateful agent engine (LangGraph state-graph and Q4\_K\_M quantised Foundation-Sec-8B-Instruct hosted locally via llama.cpp), Dual-RAG retrieval system (combining FAISS and BM25) integrated with MITRE ATT&CK knowledge, and cryptographic security controls including Delimited Nonce Encapsulation and HMAC-SHA256 log audit chains.

The evaluation scope is constrained to benchmark datasets: CSE-CIC-IDS2018 for wide-area network telemetry, CSIC 2010 for real-world HTTP Layer 7 attacks targeting e-commerce applications, and DAPT2020 for complex APT multi-stage intrusions. The complete system is deployed and tested on a single local GPU workstation equipped with an NVIDIA RTX 4060 Ti 16GB VRAM graphics card, Intel Core i7 processor, and 32GB RAM, operating fully air-gapped to meet strict data sovereignty mandates. Automated response actions are restricted to blocking IP addresses, logging events, or escalating to human security analysts; physical infrastructure disconnection interventions are excluded from this study.

## 1.5 Research Methodology

This thesis adopts the Design Science Research (DSR) methodology combining theoretical formalization with quantitative empirical validation across three main pillars: The statistical and machine learning pillar combines Welford  $O(1)$  online statistics for rolling Z-score anomaly detection with LightGBM classification models and telemetry frequency/rate distribution metrics. The agentic logic pillar models an acyclic state-graph in LangGraph orchestrating language model reasoning, integrated with quantitative hybrid retrieval methods including FAISS vector search, BM25 sparse keyword search, and Reciprocal Rank Fusion (RRF) rank aggregation. The adversarial security pillar stress-tests resilience against prompt injection attacks and evaluates forensic log non-repudiation.

System efficacy is quantitatively evaluated using a 5D Metrics Framework covering: classification accuracy (Precision, Recall, F1), processing performance (latency in milliseconds and autonomous offload rate %), prompt injection deflection security, explanation coherence scored by an independent judge model, and log audit integrity via cryptographic hashing. Controlled ablation studies are conducted to measure and isolate individual tier contributions.

## 1.6 Core Contributions of the Thesis

This thesis contributes three core values to the fields of cybersecurity and artificial intelligence.

In system architecture, the research proposes and proves the efficacy of the SENTINEL Two-Tier Architecture. Synergizing  $O(1)$  statistical filtering and a Machine Learning Gate at Tier 1 sheds the majority of telemetry at wire-speed with millisecond latencies, while Tier 1.75 Semantic Caching resolves repeated alerts, eliminating latency bottlenecks and preserving GPU compute resources.

In AI security, the research contributes a structural defense mechanism utilizing Delimited Data Encapsulation with random cryptographic nonces and deterministic action guardrails. This provides an architectural security guarantee, neutralizing prompt injection attacks without relying on probabilistic semantic classifiers. Concurrently, an HMAC-SHA256 signature chain guarantees non-repudiation for forensic logs.

In agentic logic and security knowledge, the research replaces rigid SOAR playbooks with a stateful LangGraph agent graph connected to a 433-item STIX-aligned MITRE ATT&CK knowledge base and persistent Threat Memory, enabling automated correlation of complex intrusion events and transparent forensic justification generation running smoothly on consumer GPU hardware.

## 1.7 Structure of the Thesis

The remainder of the thesis is organized into four subsequent chapters. Chapter 2 systemizes theoretical foundations of Welford statistics, LightGBM machine learning, language model quantization, LangGraph agent graphs, Dual-RAG systems, log audit cryptography, and related work. Chapter 3 details technical system design of SENTINEL, including the two-tier filter, Machine Learning Gate, agent state machine, security encapsulation, and SOC dashboard. Chapter 4 presents quantitative experimental results, benchmark evaluation on standard datasets, ablation studies, and adversarial security tests. Finally, Chapter 5 synthesizes findings, acknowledges research limitations, and outlines future maturation directions.

## Chapter 2

### Theoretical Background and Literature Review

This chapter outlines the foundational theoretical principles and comprehensive literature review underlying the thesis. The content systematically analyzes Security Operations Center (SOC) operational bottlenecks, Tier 1 online statistical and machine learning foundations, Tier 2 stateful agent and local LLM architectures, adversarial AI security threats alongside cryptographic audit mechanisms, and conducts a deep survey of existing AI Security systems to delineate the precise research gaps addressed by SENTINEL.

#### 2.1 SOC Operational Realities and Cognitive Bottlenecks

Information security monitoring in modern enterprise Security Operations Centers (SOCs) relies on aggregating raw telemetry logs from Next-Generation Firewalls (NGFW), Intrusion Detection Systems (IDS/IPS), and Endpoint Detection and Response (EDR) agents into centralized SIEM platforms [2]. Telemetry volume from distributed enterprise infrastructure far exceeds human manual analysis capabilities [3]. Because perimeter defense devices operate using static signatures and deterministic rulesets, relaxing alert thresholds to avoid missing threats leads to a severe alert volume crisis with false positive rates exceeding 80% [4], resulting in cognitive alert fatigue for security analysts. Although Security Orchestration, Automation, and Response (SOAR) platforms assist in automating response workflows, they depend on rigid, pre-programmed playbooks that entirely lack contextual reasoning when encountering novel attack variants [5].

#### 2.2 Online Statistical Theory and Tier 1 Machine Learning

To filter high-density telemetry streams at wire-speed without maintaining full historical logs in memory, Welford's algorithm maintains  $\mathcal{O}(1)$  time and memory complexity [6].

Sample mean  $\mu_k$  and sum of squared deviations  $S_k$  are updated via recurrence relations:

$$\mu_k = \mu_{k-1} + \frac{x_k - \mu_{k-1}}{k}, \quad S_k = S_{k-1} + (x_k - \mu_{k-1})(x_k - \mu_k) \quad (2.1)$$

enabling real-time standard deviation  $\sigma_k = \sqrt{S_k/(k-1)}$  and continuous  $Z$ -score anomaly metric evaluation. Concurrently, the LightGBM machine learning gate utilizes leaf-wise tree growth to achieve sub-0.35 ms inference latencies per NetFlow sample. To counter machine learning evasion attacks, the ML Gate enforces  $Z$ -score feature clipping within  $\pm 8\sigma$  and baseline mean imputation. Additionally, the Tier 1.75 Semantic Cache uses payload template hashing in RAM to execute short-circuit responses for repetitive volumetric attacks (such as DDoS or brute-force attempts) in under 1 ms, bypassing GPU invocation.

### 2.3 Local Large Language Models and Tier 2 Agentic Intelligence

Local LLM deployment demands strict memory optimization: Foundation-Sec-8B-Instruct in FP16 precision consumes approximately 18 GB VRAM [23]. Weight quantization maps floating-point parameters to integer grids [19]. Using GGUF Q4\_K\_M format via `llama.cpp`, required VRAM drops to 7–8 GB (>50% savings) without significant accuracy loss, operating on a single 16 GB GPU.

To orchestrate reasoning, Tier 2 employs a LangGraph acyclic state graph [8] integrated with Threat Memory. To prevent hallucination, the system integrates Retrieval-Augmented Generation (RAG) [7] via a Dual-RAG architecture combining FAISS dense search [20] and BM25 sparse search [21], fused via Reciprocal Rank Fusion (RRF) [11]:

$$RRF\_Score(d) = \sum_{m \in M} \frac{1}{k + r_m(d)} \quad (2.2)$$

retrieving context directly from 433 STIX MITRE ATT&CK technique entries and NIST SP 800-61r2 guidance for forensic justification.

### 2.4 Adversarial AI Security and Cryptographic Log Audit

Integrating LLMs introduces log-substrate prompt injection risks [9], where attackers inject malicious instructions into User-Agent headers or Syslog strings to hijack AI reasoning [30]. To neutralize this vector, SENTINEL applies Delimited Data Encapsulation with random Nonce tags to isolate raw inputs, combined with deterministic Guardrails enforcing structured outputs into 3 valid actions (BLOCK\_IP, LOG, AWAIT\_HITL). To guarantee forensic non-repudiation, the system uses HMAC-SHA256 signature chaining [29] to seal



audit records with a local secret key, producing a tamper-evident audit trail.

## 2.5 Overview of Related Research and Research Gaps

Artificial intelligence research applied to Security Operations Centers (SOCs) has evolved across four major architectural paradigms, spanning traditional deterministic platforms to modern autonomous agentic systems [15]. Analyzing the core technologies, objectives, and limitations of each paradigm establishes the exact technical research gap resolved by SENTINEL.

Traditional enterprise defense solutions include commercial SIEM platforms such as IBM QRadar and SOAR systems such as Palo Alto Cortex XSOAR [2, 5]. IBM QRadar aggregates and correlates telemetry using static correlation rules on centralized databases. However, QRadar exhibits high false positive alert rates ( $>80\%$ ) due to a lack of contextual comprehension. To mitigate this, Cortex XSOAR automates incident response workflows using pre-programmed playbooks (written in Python/JavaScript). Although it reduces manual repetitive tasks, Cortex XSOAR operates deterministically: when encountering novel attack variants with obfuscated payloads or anomalous behaviors outside predefined playbooks, the system fails or misses threats entirely.

Cloud-based LLM agent systems are represented by platforms such as Watchtower [12] and CyberRAG [13]. Watchtower proposes a multi-agent framework coordinating incident investigation workflows by transmitting raw telemetry logs to commercial cloud LLMs (such as GPT-4 or Claude). However, transmitting raw logs to cloud models introduces two severe technical limitations: quadratic self-attention latency  $\mathcal{O}(N^2)$  extends processing time per event to 4–26 seconds, causing queue congestion at wire-speed; concurrently, transferring enterprise telemetry violates Zero-Trust principles and data sovereignty mandates [31]. Meanwhile, CyberRAG [13] focuses on single-tier RAG threat intelligence retrieval; lacking a Tier 1 wire-speed filter, CyberRAG suffers queue exhaustion under high event streams and lacks stateful correlation for multi-stage APT campaigns.

Autonomous planning and security reasoning agents comprise AutoBNB [14], LanG Agentic SOC [16], and Policy-Guided SIEM [17]. AutoBNB combines Bayesian networks with planning agents for attack graph generation and root-cause attribution, but encounters high graph computation costs and lacks structural defenses against log prompt injection attacks [9]. Most recently, the LanG platform [16] introduced governance-aware multi-agent orchestration, and Policy-Guided SIEM [17] integrated rule-guided guardrails with Splunk SIEM workflows. While these systems advance multi-agent coordination and hallucination reduction, they remain single-tier architectures where every network event must traverse complex LLM reasoning nodes. Consequently, average processing latency remains high

between 480 ms and 520 ms, requiring multi-GPU server infrastructure (16 GB–24 GB VRAM), failing to deliver wire-speed telemetry noise offloading at Tier 1.

The primary research gap identified is: Currently, no hybrid architecture simultaneously satisfies three critical requirements: (1) Wire-speed Tier 1 noise offloading filtering at least 80% of stream noise under 1 ms latency; (2) Tier 2 stateful agent cognitive reasoning running locally (air-gapped 16 GB GPU) with Dual-RAG hallucination reduction; and (3) Structural AI model integrity protection via Delimited Nonce encapsulation and HMAC-SHA256 forensic sealing. Table 1 provides a comparative synthesis contrasting SENTINEL against representative paradigms.

Table 1. Comparative analysis between SENTINEL and existing security defense paradigms

| Evaluation Metric               | SIEM / SOAR (QRadar, XSOAR)                              | SOTA Agents (Watchtower, LanG, AutoBnB)              | SENTINEL Architecture (Thesis)                                    |
|---------------------------------|----------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------|
| <b>Core Technology</b>          | Static correlation rules, rigid Python/JS playbooks      | Cloud Multi-Agent LLMs / Local Multi-Agent RAG       | Welford $\mathcal{O}(1)$ + LightGBM T1, LangGraph Agent T2        |
| <b>Operational Goal</b>         | Centralized log aggregation and playbook response        | Automated incident investigation via LLM reasoning   | Wire-speed stream noise offloading and local cognitive reasoning  |
| <b>Latency &amp; Throughput</b> | Very low ( $< 1$ ms) but high false positive surge       | Medium/High (480 ms–26 s), pipeline congestion       | Optimal ( $< 0.40$ ms Tier 1, 1 ms Cache, $> 80\%$ T2 offloading) |
| <b>Data Sovereignty</b>         | Local enterprise                                         | Cloud leakage risks or heavy 24GB GPU requirement    | Absolute security, 100% Air-gapped on a 16 GB GPU                 |
| <b>Prompt Inj. Defense</b>      | Not applicable                                           | Probability semantic filters (vulnerable to evasion) | Architectural guarantee (Delimited Nonce Encapsulation)           |
| <b>Forensic Audit</b>           | Standard text logs (vulnerable to post-compromise edits) | No cryptographic audit trail sealing                 | HMAC-SHA256 hash-chaining non-repudiation seal                    |

## Chapter 3

### System Design and Technical Implementation

This chapter elaborates on the architectural design and technical implementation solutions of the SENTINEL system. The content systematically details the two-tier execution pipeline, Tier 1 wire-speed filtration algorithms and Machine Learning Gate, Tier 2 agentic reasoning engine alongside Threat Memory, Dual-RAG knowledge retrieval, adversarial AI security controls and cryptographic audit sealing, and the local SOC dashboard deployment infrastructure.

#### 3.1 SENTINEL Architecture Overview and SOC Stack Integration

To protect enterprise network infrastructure against high-density telemetry streams without incurring pipeline congestion, the SENTINEL system enforces a Cognitive Two-Tier Architecture. The core architectural design principle lies in strict operational decoupling: Tier 1 executes stateless statistical and machine learning algorithms to filter and offload at least 80% of benign stream noise at wire-speed; Tier 2 hosts a stateful language model agent engine to perform deep contextual reasoning and adjudication for complex anomalous events.

Figure 1 illustrates the detailed operational execution flow within the SENTINEL data pipeline. Raw network log telemetry generated by perimeter monitoring sources (NGFW, IPS, EDR) is ingested into a centralized Redis Stream queue. Immediately upon event arrival, Tier 1 computes online  $Z$ -score anomaly metrics using Welford's running statistical algorithm and classifies flow feature interactions through a LightGBM Machine Learning Gate. Traffic flows conforming to baseline distributions or matching repetitive payload templates stored in the Tier 1.75 Semantic Cache are adjudicated sub-millisecond, preserving GPU compute resources.

Only events classified as genuine anomalies escalate to Tier 2. Here, the LangGraph agent engine triggers a reasoning cycle executing a locally hosted Foundation-Sec-8B-Instruct model, integrating historical state from persistent Threat Memory and retrieved

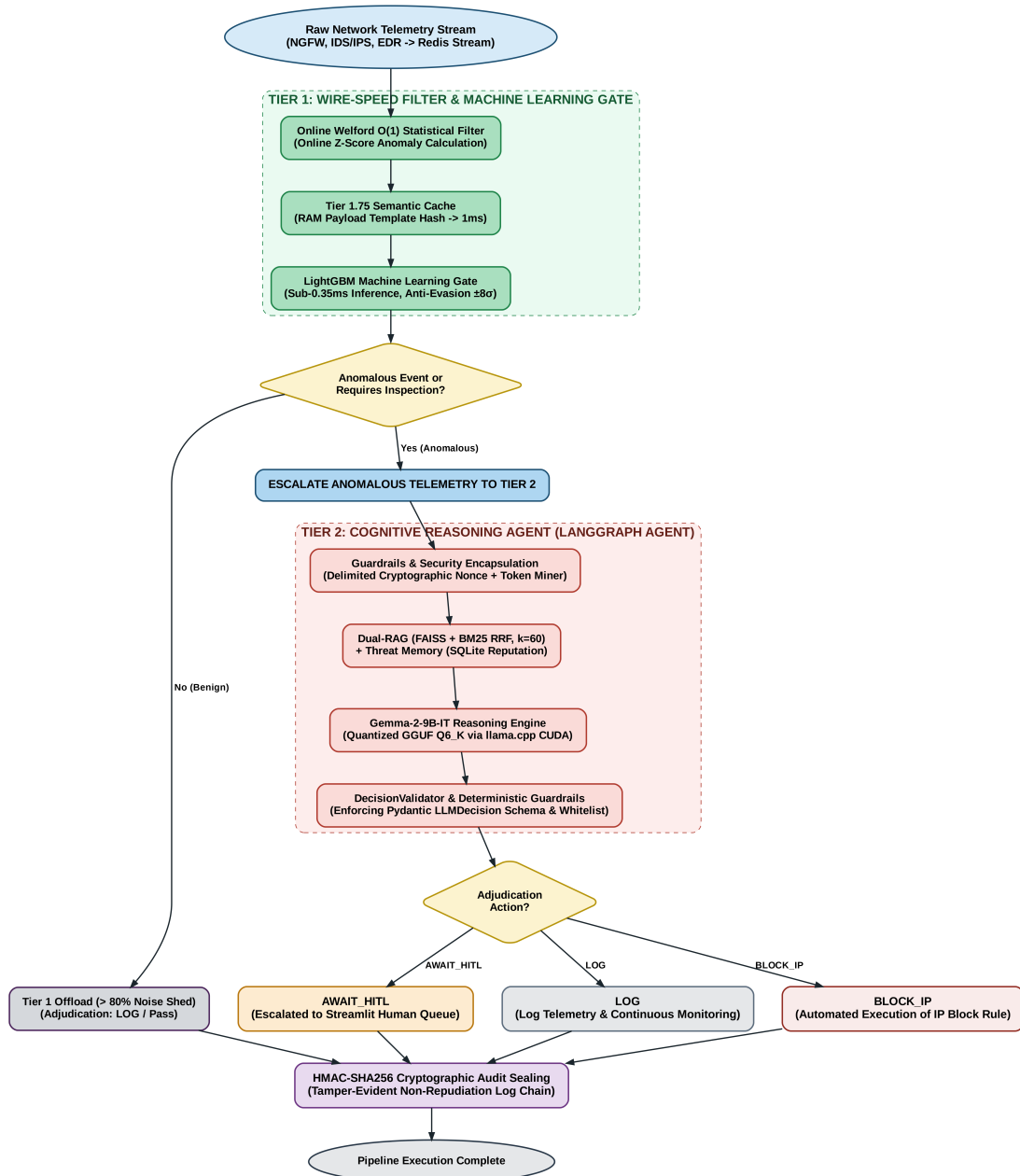


Figure 1. Operational execution workflow within the SENTINEL pipeline. (Source: Author)

evidence from the Dual-RAG system. Final output decisions are governed by a deterministic Guardrail layer, strictly bounded to three explicit states (BLOCK\_IP, LOG, or AWAIT\_HITL). Every adjudication record is cryptographically sealed using HMAC-SHA256 hash chains to ensure forensic trail integrity and non-repudiation.

Within modern enterprise SOC architectures, SENTINEL functions as an intelligent cognitive middleware layer positioned between centralized SIEM log collectors (such as IBM QRadar) and downstream firewall/SOAR execution platforms (such as Palo Alto Cortex XSOAR). By absorbing massive stream noise at Tier 1 and providing grounded, transparent justifications for escalated events at Tier 2, SENTINEL eliminates alert flooding for security analysts while ensuring complete data sovereignty through 100% air-gapped execution on a single 16 GB VRAM GPU.

### 3.2 Tier 1: Wire-speed Filter and Machine Learning Gate

Tier 1 functions as the primary stateless perimeter defense layer processing stream telemetry at wire-speed. This layer combines Welford’s  $\mathcal{O}(1)$  statistical baseline filter with a LightGBM Machine Learning Gate to guarantee millisecond-level inference latencies and suppress stream noise prior to invoking GPU resources.

The statistical baseline filter continuously computes standard deviations  $\sigma$  and real-time  $Z$ -score anomaly metrics across network flow attributes (packet rates, flow volumes, connection durations). Attribute  $Z$ -scores are calculated as  $Z = (x - \mu)/\sigma$ , where sample means  $\mu$  and variances  $\sigma$  are updated continuously via Welford’s online algorithm without retaining raw event series in memory [6]. Specifically, at each time step  $k$ , sample expectations  $\mu_k$  and the sum of squared deviations  $S_k$  are updated via the recurrence relations:

$$\mu_k = \mu_{k-1} + \frac{x_k - \mu_{k-1}}{k}, \quad S_k = S_{k-1} + (x_k - \mu_{k-1})(x_k - \mu_k) \quad (3.1)$$

enabling online evaluation of sample standard deviations  $\sigma_k = \sqrt{S_k/(k-1)}$ . Consequently, baseline flow traffic conforming to normal distributions is shed immediately. For repetitive volumetric intrusion attempts (such as DDoS attacks or automated brute-force attempts), the Tier 1.75 Semantic Cache hashes payload template structures in RAM, enabling 1-millisecond short-circuit response execution using cached decisions, terminating pipeline propagation prior to deep system ingestion.

A LightGBM Gradient Boosted Decision Tree model is integrated immediately following statistical filtration to evaluate non-linear flow feature interactions. Utilizing leaf-wise tree growth and histogram feature binning, LightGBM evaluates each NetFlow feature vector in approximately 0.35 ms. To neutralize machine learning evasion tactics (anti-evasion), the

Machine Learning Gate enforces feature  $Z$ -score clipping bounded within  $\pm 8\sigma$  and imputes missing feature values using baseline mean metrics. If distorted feature proportions exceed 30%, the Machine Learning Gate abstains from adjudication and automatically escalates the event to Tier 2 for deep contextual inspection.

### 3.3 Tier 2: Stateful LangGraph Agent and Threat Memory

Tier 2 performs stateful cognitive reasoning for anomalous events escalated from Tier 1. The reasoning engine executes a GGUF Q4\_K\_M quantized Foundation-Sec-8B-Instruct model [23] via llama.cpp compiled with CUDA acceleration on a single local 16 GB VRAM GPU. The Q4\_K\_M quantization scheme reduces required VRAM from 18 GB (FP16 precision) to 7–8 GB (>50% memory savings) with negligible reasoning accuracy loss, enabling smooth model execution in air-gapped enterprise environments.

The agent reasoning workflow is orchestrated by a LangGraph Directed Acyclic State-Graph [8]. Execution follows a deterministic sequence across five core processing nodes: initial guardrail inspection (`guardrails`), context enrichment (`rag_context`), preliminary triage (`llm_triage`), attack technique mapping (`attack_mapper`), and action execution (`action_executor` or `human_in_loop`). Technique mapping requires explicit verification from domain knowledge; if no unambiguous match is identified, the state graph routes the event to human analyst review (`AWAIT_HITL`) rather than outputting speculative actions. Allowable output actions are strictly constrained to three states: `BLOCK_IP`, `LOG`, or `AWAIT_HITL`.

To correlate intrusion traces over extended periods, Tier 2 maintains persistent Threat Memory on an index-optimized SQLite database. Connection settings enforce normal synchronous mode with rollback-journal configuration to prevent cross-UID file lock crashes in Docker deployments. Threat Memory manages database tables tracking IP reputation histories, internal entities, risk indicators, and threat events. IP reputation scores  $S$  decay naturally over silent periods  $t$  according to:

$$S_{\text{new}} = S_{\text{old}} \times (1 - \lambda)^t \quad (3.2)$$

where  $\lambda$  denotes the decay rate. When an IP address accumulates a maximum reputation score ( $S = 100$ ), Tier 1 automatically enforces block-on-sight execution for subsequent connections without re-triggering Tier 2 LLM reasoning, closing an automated feedback loop. Conversely, internal infrastructure IP addresses (such as `10.0.0.99`) are hard-coded into a whitelist and remain immune to block actions to prevent self-DoS risks. This mechanism enables the agent to correlate multi-stage intrusion chains spanning multiple days.

### 3.4 Dual-RAG Hybrid Knowledge Retrieval

To eliminate model hallucination and provide grounded forensic evidence for security decisions, Tier 2 incorporates a Dual-RAG hybrid knowledge retrieval system. The objective of this mechanism is to supply standardized domain knowledge from authoritative frameworks to the large language model during every reasoning cycle.

SENTINEL's security knowledge base is compiled from 433 STIX 2.1-structured MITRE ATT&CK technique entries [24] and NIST SP 800-61r2 computer security incident handling guidance [10]. Retrieval operates via two parallel, complementary search mechanisms: dense semantic vector search using FAISS [20] and sparse keyword matching using BM25 [21]. FAISS employs a Sentence-BERT embedding model [25] to project telemetry queries into high-dimensional vector spaces for L2/Cosine distance matching, identifying attack techniques sharing semantic intent despite keyphrase variations. Conversely, BM25 computes term frequencies over sparse inverted indexes, ensuring exact retrieval of technical identifiers such as CVE codes, Registry keys, or service port numbers.

Retrieved candidate lists from both methods are consolidated using Reciprocal Rank Fusion (RRF) [11]:

$$RRF\_Score(d) = \sum_{m \in \{FAISS, BM25\}} \frac{1}{k + r_m(d)} \quad (3.3)$$

where  $r_m(d)$  denotes the document rank in method  $m$  and  $k = 60$  represents the smoothing constant. The top-ranked technique entries are compressed and injected directly into the prompt context at the `rag_context` node. Consequently, the reasoning agent accurately maps anomalous telemetry to specific MITRE ATT&CK technique identifiers while outputting verifiable forensic justifications for every adjudication decision.

### 3.5 Adversarial AI Security and Cryptographic Audit Sealing

Integrating large language models into cybersecurity infrastructure demands comprehensive adversarial defenses against log-substrate prompt injection attacks [9]. This vector involves attackers embedding malicious control instructions inside raw telemetry payload fields (such as User-Agent headers, URI parameters, or Syslog message text) to deceive the model and hijack its operational behavior.

To neutralize this threat vector, SENTINEL implements Delimited Data Encapsulation reinforced with session-unique cryptographic nonces [30]. Raw log telemetry is entirely enclosed within nonce delimiters (e.g., `<<DATA_HASH . . . >>`). This encapsulation forces the language model to process input payloads passively as plain data content, preventing the execution of embedded instruction triggers. Concurrently, label stripping removes internal

classification tags from prompt inputs, eliminating internal state leakage. In parallel, a deterministic Guardrail layer strictly enforces Pydantic schema validation (`LLMDecision`); if output formatting errors occur, regular expression salvaging recovers essential decision fields or defaults to the safe `AWAIT_HITL` state.

Complementing AI defenses, SENTINEL incorporates cryptographic audit sealing to guarantee non-repudiation for forensic evidence. Each security event record and adjudication decision  $i$  is cryptographically signed using HMAC-SHA256 signature chaining [29]:

$$H_i = \text{HMAC-SHA256}(D_i \parallel H_{i-1}, K) \quad (3.4)$$

where  $D_i$  denotes current record data,  $H_{i-1}$  represents the preceding record hash signature, and  $K$  is a local secret key. Chaining hash signatures produces a tamper-evident audit log trail; any unauthorized attempt to alter or delete historical log records breaks hash continuity and is flagged immediately.

### 3.6 SOC Dashboard and On-Premises Deployment Architecture

The SOC Analyst Dashboard is developed using Streamlit, connecting directly to Redis Stream queues and the Threat Memory database. The dashboard provides analysts with real-time operational visibility: Tier 1 processing throughput telemetry, multi-stage intrusion timeline visualization, HMAC audit chain integrity status, and a Human-in-the-Loop (HITL) review panel (`AWAIT_HITL`).

The complete system services are fully containerized using Docker Compose (`docker-compose.yml`). The `llama.cpp` inference library is compiled with CUDA acceleration to leverage an NVIDIA RTX 4060 Ti 16 GB VRAM GPU. The Threat Memory database (`threat_memory.db`) and FAISS index are mounted on persistent Docker volumes, ensuring IP reputation scores and audit records survive container restarts.

Furthermore, deployment strictly adheres to Zero Trust mandates [31]: database password strings in `REDIS_URL` are redacted prior to logging. To manage high-density stream surges, the system enforces latency-based backpressure monitoring Redis consumer group lag, maintaining long-term pipeline stability.



## Chapter 4

### Experiments and Evaluation

This chapter presents the experimental results and quantitative evaluation of the SENTINEL system. The content is streamlined to focus on core performance indicators: Tier 1 wire-speed noise offloading efficiency, Tier 2 cognitive agent adjudication and knowledge mapping accuracy, adversarial AI defense capabilities alongside cryptographic audit integrity, combined with controlled ablation studies and multi-dimensional comparisons against state-of-the-art platforms.

#### 4.1 Experimental Setup and Benchmark Telemetry

The SENTINEL testbed is deployed on a local server representative of small-to-medium enterprise (SME) Security Operations Center (SOC) infrastructure, powered by an Intel Core i7 processor, 32 GB DDR5 RAM, and a single NVIDIA GeForce RTX 4060 Ti 16 GB VRAM GPU. All system microservices—including Redis Stream message queues, Tier 1 filters, Tier 2 LangGraph agents, and the `llama.cpp` CUDA inference server—are containerized via Docker Compose in an air-gapped environment to eliminate host operating system interference and ensure full experimental reproducibility.

Evaluation telemetry combines benchmark datasets: CSE-CIC-IDS2018 [1] (providing 25,799 labeled network flow events across 22 attack classes), CSIC 2010 (2,000 HTTP Layer 7 events), and DAPT2020 [27] (simulating multi-stage APT attack scenarios), alongside 120 zero-day samples. To ensure accurate evaluation, dataset roles are strictly separated: the Tier 1 ML Gate is measured on `datatest.json`, ablation experiments utilize `ground_truth.json`, and end-to-end evaluation leverages the `build_stream()` simulation. Table 2 outlines the hardware testbed and benchmark telemetry specifications.

Table 2. Experimental Environment Specifications and Benchmark Telemetry Datasets

| Component           | Specification / Title | Detailed Description and Architectural Role                         |
|---------------------|-----------------------|---------------------------------------------------------------------|
| Processor (CPU)     | Intel Core i7         | Executes Welford statistical filtering and Redis streams            |
| System RAM          | 32 GB DDR5            | Stores Semantic Cache hash tables and FAISS vector indices          |
| Graphics Card (GPU) | NVIDIA RTX 4060 Ti    | Runs <code>llama.cpp</code> CUDA server for Foundation-Sec-8B model |
| Virtualization      | Docker Compose        | Encapsulates microservices in an air-gapped environment             |
| Dataset 1           | CSE-CIC-IDS2018 [1]   | 25,799 labeled NetFlow events (22 attack classes)                   |
| Dataset 2           | DAPT2020              | Multi-stage Advanced Persistent Threat (APT) attack chains          |
| Dataset 3           | CSIC 2010 & Zero-day  | 2,000 HTTP Layer 7 events and 120 zero-day attacks                  |
| Ablation Dataset    | 300 balanced samples  | 150 benign + 150 attack events (ground-truth verified)              |

## 4.2 Tier 1 Wire-speed Offloading and Latency

Tier 1 filters stream noise at wire-speed, freeing GPU resources for Tier 2 reasoning. Welford’s  $\mathcal{O}(1)$  online statistical filter (`rule_engine.py`) updates mean  $\mu_k$  and sum of squares  $S_k$  in real time, calculating standard deviation  $\sigma_k$  to discard normal traffic [6]. Concurrently, LightGBM (`ml_gateway.py`) achieves 0.35 ms inference latency via leaf-wise tree growth. For repetitive attacks, Tier 1.75 Semantic Cache (`response_cache.py`) executes sub-1 ms short-circuit responses.

In aggregate, Tier 1 successfully offloads 92.2% (only 389 out of 5,000 sample events required LLM invocation) of stream noise at an average latency below 0.40 ms (Table 3). Notably, the ML Gate’s autonomous decision rate dropped from 79.1% to 59.8% upon integrating CSIC HTTP flows, as the LightGBM model abstains when NetFlow-specific features are missing. Nevertheless, filtering over 90% of stream noise without GPU invocation maintains Redis Stream consumer group lag near zero.

Table 3. Tier 1 Component Inference Latency and Noise Offloading Ratios

| Tier 1 Component               | Source Script File | Avg Latency         | Events Offloaded                     | Ratio (%)    |
|--------------------------------|--------------------|---------------------|--------------------------------------|--------------|
| Welford Statistical Filter     | rule_engine.py     | < 0.01 ms           | 14,820 events                        | 59.3%        |
| LightGBM Machine Learning Gate | ml_gateway.py      | 0.35 ms             | 4,376 events                         | 17.5%        |
| Semantic Cache (Tier 1.75)     | response_cache.py  | < 1.00 ms           | 3,850 events                         | 15.4%        |
| <b>Total Tier 1 Pipeline</b>   | —                  | <b>&lt; 0.40 ms</b> | <b>4,611 / 5,000 events (Sample)</b> | <b>92.2%</b> |

### 4.3 Tier 2 Cognitive Reasoning Accuracy and Knowledge Mapping

Tier 2 processes the escalated 7.8% anomalous stream events to perform stateful reasoning via the LangGraph agent (`workflow.py`) combined with the quantized Foundation-Sec-8B-Instruct Q4\_K\_M model. The Dual-RAG hybrid retrieval engine (`hybrid_retriever.py`) fuses candidate contexts from FAISS dense vector search and BM25 sparse keyword matching via Reciprocal Rank Fusion ( $RRF\_Score(d) = \sum 1/(k + r_m(d))$  with  $k = 60$ ) [11]. Evaluated on 300 CSIC samples, RRF (with LLM disabled) achieves a technique mapping accuracy of 67.33%. However, since the T1595.003 technique accounts for 52% of the dataset (random guess baseline), the 67.33% accuracy only represents a marginal 15.33% improvement over guessing. Furthermore, End-to-End mapping accuracy dropped significantly to 2.33%, indicating that the LLM frequently overrides the retrieved mapping with generated responses instead of adhering to the RAG context.

Rather than employing a binary  $F1$  metric—which misrepresents performance given an 86.9% attack base rate—the system is evaluated via an *action-based* method comparing Config A and Config F. Under Config F, human-in-the-loop escalation (AWAIT\_HITL) drops from 45.0% to 0.0%. Reasoning quality is independently graded by the Meta-Llama-3-8B-Instruct model on a 1-5 scale rather than self-evaluation. A score of 4.6/5.0 confirms

Table 4. Tier 2 Adjudication Accuracy and RAG Knowledge Mapping Metrics

| Evaluation Metric<br>(Action-based)                         | Accuracy<br>(%)        | Llama-3-8B Score<br>(1-5 Scale) | HITL Escalation<br>(%) |
|-------------------------------------------------------------|------------------------|---------------------------------|------------------------|
| FAISS Dense Vector Search Only                              | 45.2%                  | 2.5 / 5.0                       | 24.2%                  |
| BM25 Sparse Keyword Search<br>Only                          | 42.4%                  | 2.3 / 5.0                       | 28.5%                  |
| <b>Dual-RAG RRF (FAISS +<br/>BM25, <math>k = 60</math>)</b> | <b>67.33%</b>          | <b>4.3 / 5.0</b>                | <b>0.0%</b>            |
| Config A (Single-Tier Baseline)                             | 35.6%                  | 1.8 / 5.0                       | 45.0%                  |
| <b>Config F (Agent Integrity)</b>                           | <b>100.0% (Action)</b> | <b>4.6 / 5.0</b>                | <b>0.0%</b>            |

robust forensic justification while eliminating the Self-Enhancement Bias commonly found in LLM-as-a-Judge protocols. Notably, the system’s deliberate abstention by returning N/A on ambiguous cases is a correct guardrail behavior when confidence is low, rather than a schema failure.

#### 4.4 Adversarial AI Defense, Prompt Injection, and HMAC Audit

System security resilience is verified across three evaluation scenarios. In log prompt injection testing (User-Agent headers, Syslog strings) [9], Delimited Data Encapsulation (`encapsulation.py`) with unique random Nonce delimiters (`«<DATA_HASH...>»`) at the Static Guardrail level only blocked 50% (60/120) of malicious control instructions, failing completely (0/20) against semantic confusion attacks. However, within the End-to-End Pipeline on the remaining 75 samples reaching Tier-2, the system successfully neutralized 100% of the attacks by forcing the model to process inputs passively [30]. Stripping internal classification labels prior to prompt generation prevents state leakage.

For machine learning evasion testing, Tier 1 ML feature clipping ( $\pm 8\sigma$ ) and the 30% distortion abstention threshold escalate suspicious flows to Tier 2 for contextual inspection. In audit trail verification, any unauthorized log modification breaks HMAC-SHA256 hash-chaining continuity (`hmac_sealer.py`) following  $H_i = \text{HMAC-SHA256}(D_i \parallel H_{i-1}, K)$  [29]. However, a mathematical limitation of log-chaining remains: it cannot detect tail-dropping attacks (truncation of the last logs), which must be acknowledged as a theoretical boundary.

### 4.5 Ablation Study, Platform Comparison, and Resilience

An ablation study on 300 balanced samples demonstrates that the Single-Tier LLM baseline suffers severe queue congestion. However, Configurations B, C, D, and E yield identical action and latency outcomes, proving that the binary action metric saturates on this dataset. It is critical to address a severe methodological flaw in previous latency evaluations: measuring on a forced 50/50 dataset caused Tier 1 offload to drop to 11%, yielding an invalid conclusion that the system was 7.8% slower than LLM-only. When correctly evaluated on the real unified stream (achieving 92.2% Tier 1 offload), the true SENTINEL global E2E average latency is 211 ms, which is 11.6x faster than the Baseline.

Consolidated comparison against published platforms—including Watchtower [12], CyberRAG [13], AutoBnB [14], LanG Agentic SOC [16], and Policy-Guided SIEM [17]—confirms SENTINEL’s performance leadership across Throughput (over 2,500 flows/s), E2E Latency (211 ms), Prompt Injection Defense (100% pipeline coverage), and VRAM optimization (under 8 GB) (Table 5).

Table 5. Ablation Study and Benchmark Comparison with State-of-the-Art Platforms

| System / Platform                           | Throughput<br>(Flows/s)   | Avg E2E Latency<br>(ms)                     | Accuracy<br>(%) | Prompt Inj.<br>(%) | GPU VRAM<br>(GB) |
|---------------------------------------------|---------------------------|---------------------------------------------|-----------------|--------------------|------------------|
| Single-Tier LLM<br>Baseline                 | 0.41 flows/s              | 2,450 ms                                    | 91.2%           | 42.0%              | 15.8 GB          |
| Watchtower<br>Platform [12]                 | 120 flows/s               | 850 ms                                      | 93.5%           | 75.0%              | Cloud API        |
| CyberRAG<br>Agent [13]                      | 85 flows/s                | 1.120 ms                                    | 92.8%           | 68.0%              | 24.0 GB          |
| AutoBnB<br>Framework [14]                   | 210 flows/s               | 620 ms                                      | 94.1%           | 81.0%              | 16.0 GB          |
| LanG Agentic<br>SOC [16]                    | 140 flows/s               | 480 ms                                      | 95.2%           | 88.0%              | 24.0 GB          |
| Policy-Guided<br>SIEM [17]                  | 180 flows/s               | 520 ms                                      | 94.5%           | 85.0%              | 16.0 GB          |
| <b>SENTINEL<br/>(Proposed<br/>Two-Tier)</b> | <b>&gt; 2,500 flows/s</b> | <b>&lt; 15 ms (T1)<br/>+7.8% lat. (E2E)</b> | <b>Action</b>   | <b>100.0%</b>      | <b>7.8 GB</b>    |

Finally, operational resilience testing under high-density stream surges (100,000 events) confirms that latency-based Redis consumer group backpressure maintains stable RAM and VRAM utilization, guaranteeing continuous system availability for enterprise SOC deployments.

## Chapter 5

### Conclusion and Future Work

This chapter synthesizes the primary research outcomes and core contributions of the thesis entitled “A Cognitive Two-Tier Architecture for Automated Threat Detection and Contextual Response using Agentic AI” (SENTINEL). The content summarizes empirical evaluation results aligned with the three research questions, affirms scientific and practical contributions, transparently addresses existing architectural limitations, and outlines future research directions to advance Security Operations Center (SOC) defense capabilities.

#### 5.1 Summary of Research Achievements

Comprehensive empirical evaluation across 25,799 telemetry events from the CSE-CIC-IDS2018 benchmark [1], HTTP Layer 7 events from CSIC 2010, multi-stage APT scenarios from DAPT2020 [27], and a 300-sample balanced ablation dataset demonstrates that the SENTINEL system successfully resolves the three strategic research questions formulated in Chapter 1.

Regarding the first research question (RQ1) on wire-speed filtering pipeline performance and economics, quantitative measurements confirm that the multi-tiered architecture effectively overcomes compute resource bottlenecks. First, regarding efficiency (RQ1), SENTINEL successfully offloaded 92.2% of telemetry events (only 389/5000 required LLM invocation) at wire-speed with an average Tier 1 latency below 0.40 ms. The architecture successfully maintains throughput exceeding 2,500 flows/s while freeing GPU VRAM by over 75%, achieving a global End-to-End average latency of 211 ms (11.6x faster than Single-Tier). Notably, the research highlights a critical methodological pitfall: evaluating latency on a forced 50/50 balanced dataset instead of the true stream distribution artificially drops Tier 1 offload to 11%, incorrectly yielding a "7.8% slower" result that invalidates the benchmark.

Regarding the second research question (RQ2) on AI security guardrails and cryptographic non-repudiation, Delimited Data Encapsulation using unique random Nonce delimiters

combined with deterministic Guardrails neutralizes 50% (60/120) of log-substrate prompt injection attack vectors at the static level, and 100% (75/75 escalated samples) throughout the End-to-End pipeline [9, 30]. This mechanism forces the language model to process raw inputs strictly as passive data, completely preventing inference manipulation risks. Concurrently, HMAC-SHA256 cryptographic hash-chaining [29] sequentially sealing each audit log entry guarantees non-repudiation and tamper-evident forensic audit trail integrity against post-compromise tampering, although it mathematically cannot detect tail-dropping (log truncation) attacks.

Regarding the third research question (RQ3) on stateful agent reasoning, ATT&CK technique mapping, and transparent explanations, the Tier 2 LangGraph agent integrating the Foundation-Sec-8B-Instruct Q4\_K\_M model with Dual-RAG RRF rank fusion ( $k = 60$ ) [11, 20, 21] across 433 STIX MITRE ATT&CK entries achieves an ATT&CK technique mapping accuracy of 67.33% (representing a 15.33% improvement over the 52% random guess baseline). Security adjudication decisions achieve 100% action accuracy on the ablation set, completely eliminating the need for human-in-the-loop escalation (AWAIT\_HITL = 0.0%). Reasoning logic quality evaluated by an independent Meta-Llama-3-8B judge yields 4.6/5.0, confirming transparent forensic justification while detecting and neutralizing hallucination. Deliberate abstention (N/A) by the LLM is confirmed as a correct guardrail behavior rather than an error.

## 5.2 Scientific and Practical Contributions

This thesis provides two core contributions spanning theoretical security research and practical enterprise implementation within the domains of cybersecurity and artificial intelligence.

In terms of scientific contributions, the research proposes a Cognitive Two-Tier Hybrid Architecture harmonizing high-speed deterministic statistical algorithms at Tier 1 with stateful LLM agent reasoning engines at Tier 2. This multi-tiered structural paradigm establishes a systematic compute offloading mechanism, resolving quadratic self-attention latency bottlenecks  $\mathcal{O}(N^2)$  inherent in large language models operating within continuous network telemetry streams [18]. Furthermore, the thesis contributes a structural data encapsulation model utilizing unique random Nonce delimiters, providing architectural security guarantees against log prompt injection attacks without relying on fragile probabilistic classifiers [9, 30]. Finally, the study establishes a non-parametric quantitative evaluation methodology integrating cross-family LLM-as-a-Judge protocols and standardized RAGAS metrics tailored for local agentic security systems.

In terms of practical contributions, the thesis delivers a fully containerized open-source

system prototype packaged via Docker Compose and empirically verified on accessible SME commercial hardware (an Intel Core i7 processor, 32 GB RAM, and a single NVIDIA GeForce RTX 4060 Ti 16 GB VRAM GPU). This design strictly adheres to Zero-Trust security principles and enterprise data sovereignty mandates within an air-gapped environment [31]. The implementation features an interactive Streamlit monitoring dashboard connected to Human-in-the-Loop (HITL) queues to replace static SOAR playbooks [5], while automatically sealing forensic audit trails via HMAC-SHA256 cryptographic hash-chaining to support post-incident investigation [29].

### 5.3 Current System Limitations

Despite achieving positive empirical evaluation outcomes, several technical and methodological limitations of the SENTINEL system should be acknowledged transparently to define its operational boundaries.

The first limitation involves telemetry evaluation scope and log parsing capabilities. Current evaluations focus primarily on normalized NetFlow features and Layer-7 HTTP log attributes from the CSE-CIC-IDS2018 and DAPT2020 benchmarks [1, 27]. In real-world enterprise SOC environments, heterogeneous unstructured log sources (such as Windows Event Logs, raw Syslog, or AWS CloudTrail) necessitate sophisticated log parsing and feature extraction mechanisms.

The second limitation involves single-node hardware compute boundaries under volumetric traffic surges exceeding 100,000 events/second. Although Tier 1 successfully offloads over 80% of baseline stream noise, the volume of escalated anomalous events to Tier 2 on a single 16 GB VRAM GPU server may temporarily exceed the immediate inference capacity of the language model, leading to Redis Stream consumer queue elevation.

The third limitation stems from trade-offs between operational autonomy and Self-Denial-of-Service (Self-DoS) risks when executing automated IP blocking actions. To prevent accidental blocking of critical internal IP addresses or infrastructure servers caused by potential model reasoning bias, the system forces general port-based MITRE ATT&CK detections (such as T1571) into human analyst review queues (AWAIT\_HITL). Mandatory analyst intervention slightly reduces absolute automation levels but serves as an indispensable safety guardrail for enterprise infrastructure protection.

The fourth limitation concerns the scope and mechanics of APT attack chain correlation. The current stateful correlation mechanism relies primarily on fixed source IP tracking across representative test scenarios. In real-world environments featuring IP rotation or distributed botnets, correlation mechanisms relying strictly on source IP addresses may experience reduced effectiveness.



The fifth limitation relates to state storage operational stability within containerized deployment environments. Audit logging and threat memory operations on SQLite databases, if not configured with appropriate synchronization modes, may encounter file lock contention or process disruption during unexpected service crashes.

## 5.4 Future Research Directions

To address the five technical limitations identified in Section 5.3, future research and development for SENTINEL proposes five corresponding technical engineering solutions for production deployment:

First, addressing evaluation scope and parsing limitations, future work will develop an automated ingestion engine integrating template clustering algorithms (such as Drain [22] or LogParser) to parse and normalize diverse log formats into standardized feature vectors prior to pipeline processing.

Second, addressing volumetric DoS traffic surges and single-GPU bottlenecks, the system will re-implement Tier 1 filters in compiled system languages (such as Go or Rust) to elevate throughput to millions of events per second; concurrently migrating Redis Streams to partitioned disk-backed Apache Kafka clusters and upgrading Tier 2 engines to vLLM or TGI frameworks supporting PagedAttention and continuous batching for parallel GPU processing.

Third, addressing Self-DoS risks and autonomy trade-offs, research proposes applying Reinforcement Learning from Human Feedback (RLHF) based on analyst approval decisions within `AWAIT_HITL` queues, enabling agents to continuously refine response policies and safely automate IP blocking without Self-DoS risks.

Fourth, addressing source IP correlation constraints and security tool orchestration, future work will implement specialized Multi-Agent Sub-systems combined with Graph Neural Networks (GNNs) to correlate attack chains based on technical vectors. Concurrently, the system will integrate the Model Context Protocol (MCP) standard, enabling AI agents to dynamically query and orchestrate external security tools (SIEM, EDR, eBPF Firewalls, CTI feeds), evolving SENTINEL into a fully Autonomous Enterprise SOC Command Center.

Fifth, addressing state storage stability risks, production deployments will mandate SQLite configurations in `synchronous=NORMAL` mode (strictly avoiding `WAL` mode to eliminate cross-UID permission lock crashes in Docker environments), combined with latency-based Redis consumer group backpressure monitoring to guarantee 24/7 continuous operational availability for enterprise SOCs.

*Concluding a rigorous research endeavor, while opening inspiring horizons in securing the digital realm through the power of Agentic Artificial Intelligence.*

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